Name: _____ Roll No.: _____

EE 2001 Digital Systems Final Exam

Monday, 06/05/2024

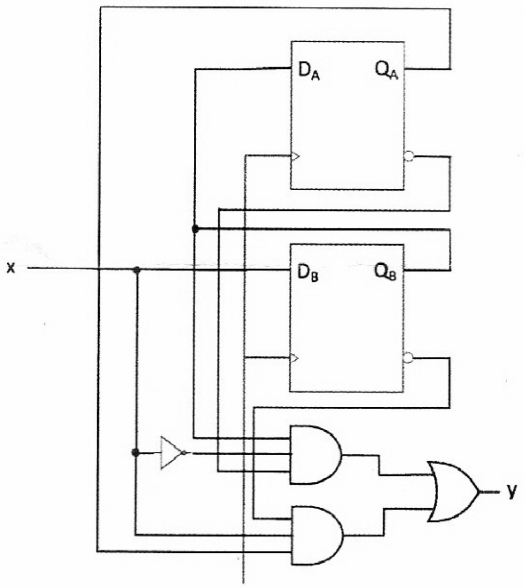
Closed book, 180 mins, 30 points

Remember...

- *Make reasonable assumptions when in doubt. However, you should show all steps and justify assumptions for full credit.*

1. Design and implement a combinational logic circuit that performs a matrix multiplication of a 1x2 matrix consisting of two bit binary numbers and a similar 2x1 matrix. Determine the minimum number of half-adders, full-adders, and other gates required to perform the above task. (5 pts)

2. Analyze the following synchronous sequential circuit with the help of a state table and a corresponding state diagram. Identify whether it is a Mealy or Moore-type circuit and also the operation that it performs. What will be the output for the input sequence 0010101101000 (read from left to right)? (5 pts)



3. Draw the state diagram, state table, and realize a Moore-type error detection circuit for a binary communication link using J-K flip-flops and additional gates, if necessary. When the communication link operates correctly, all blocks of 0's are of odd length and all blocks of 1's are of even length. Whenever an error occurs, i.e. a block of 0's contains an even number of 0's or a block of 1's contain an odd number of 1's, the output Z = 1 immediately with change in input. At all other instances, Z = 0. For example if the input (read from left to right) is 0001100111011110...., the output is 0000000100100000..... (5 pts)

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4. (a) The contents of a four-bit register are initially 1101. The register is shifted six times to the right with the serial input being 1010111. What is the content of the register after each shift? (2 pts)

(b) Design a synchronous counter with the following repeated binary sequence: 0, 1, 2, 4, 6. Implement the counter using D flip-flops. (3 pts)

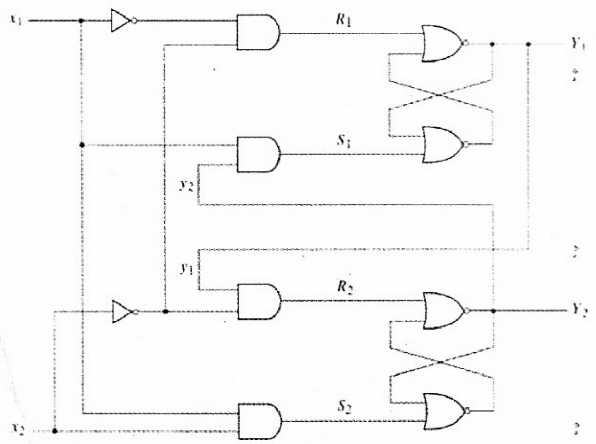
5. (a) Implement the following two Boolean functions using a PLA circuit with minimum number of AND gates and show the corresponding programming table. (2.5 pts)

$$F1(A, B, C) = \sum (0, 1, 2, 4)$$

$$F2(A, B, C) = \sum (0, 5, 6, 7)$$

(b) Assume there are two sensors S1 and S2 in a tank at levels L1 and L2 respectively (L1<L2). There are three lights, Red (R), Orange (O), and Green (G), and a motor control as outputs. The motor should stop after it reaches L2, and the light Red (R) must glow when the water level is below L1, Orange (O) should glow when the water level is between L1 and L2, and Green (G) should glow when the water level is at (or above) L2. Write the truth table for this operation and implement it using PAL. (Note that sensors will be ON when the water level is touching them). (2.5 pts)

6. (a) For the asynchronous sequential circuit shown below, (i) derive the transition table and flow table of the circuit, (ii) Determine if the circuit is stable or not, (iii) Determine all race conditions and check whether they are critical or non-critical. (3 pts)



(b) Draw the logic diagram of the product-of-sums expression

$$Y = (X1 + X2') (X2 + X3)$$

Show that there is a static 0-hazard when X1 and X3 are equal to 0 and X2 goes from 0 to 1. Find a way to remove the hazard. (2 pts)